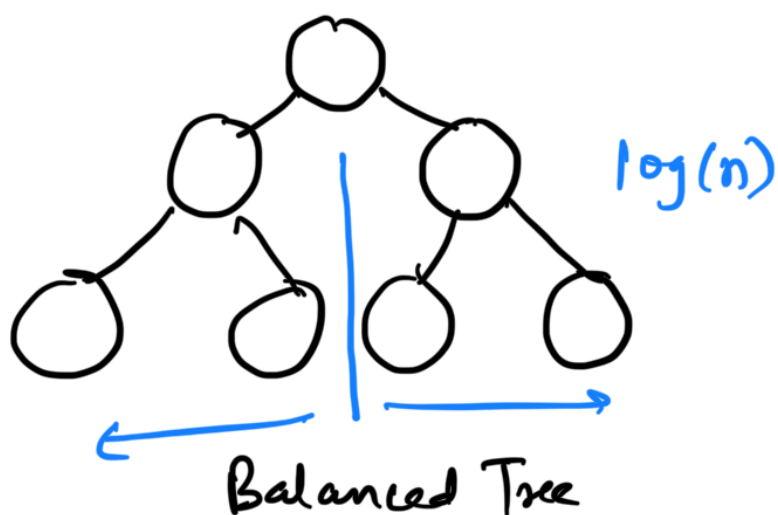
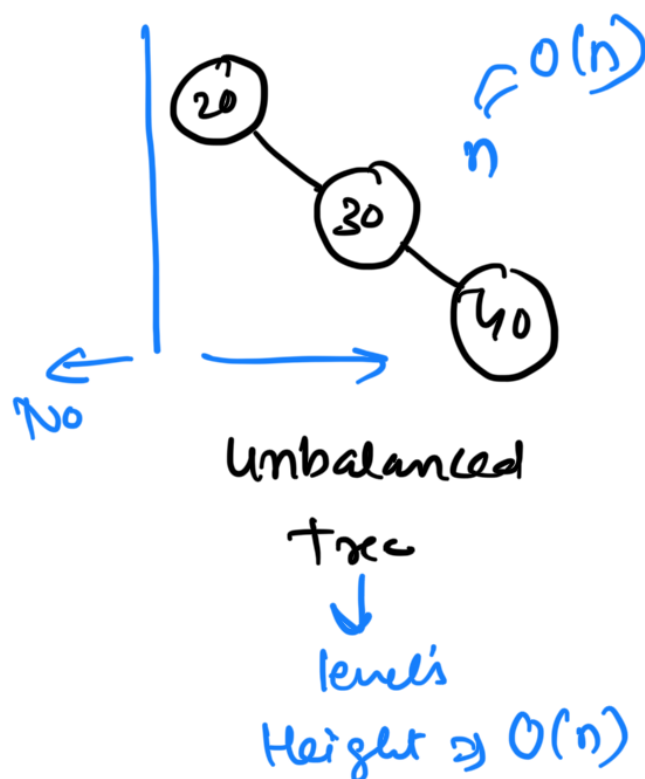
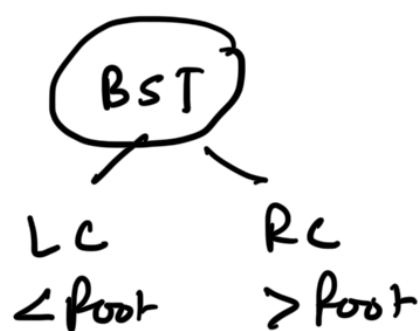


Tree Applications

Balanced Search Tree



Height $\Rightarrow$ level
 $= \log n$



Balanced Tree

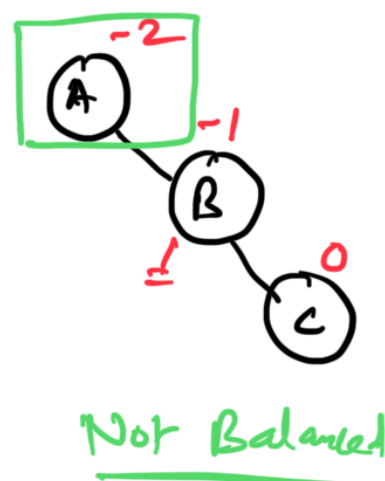
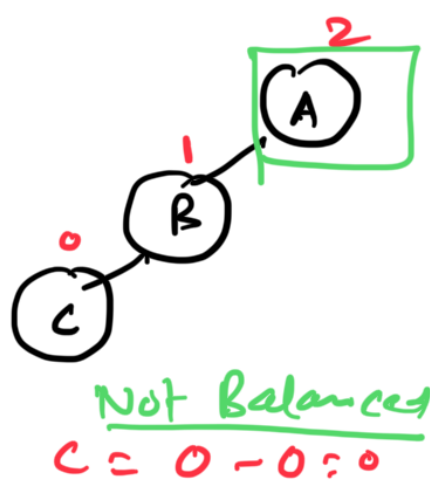
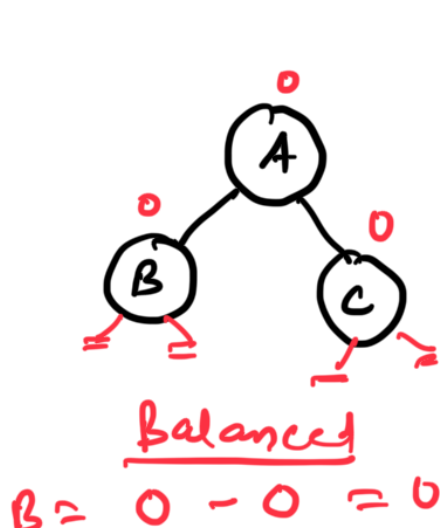
1. AVL Tree
2. Red-Black Tree
3. B-Trees, n any trees

AVL Tree $\rightarrow$ BT $\rightarrow$ BST

- Adelson-Velski - Landis
- Balance Factor

$$BF = h(\text{left subtree}) - h(\text{right subtree})$$

BF $\in \{-1, 0, +1\}$



$$C = 0 - 0 = 0$$

$$B = 1 - 0 = 1$$

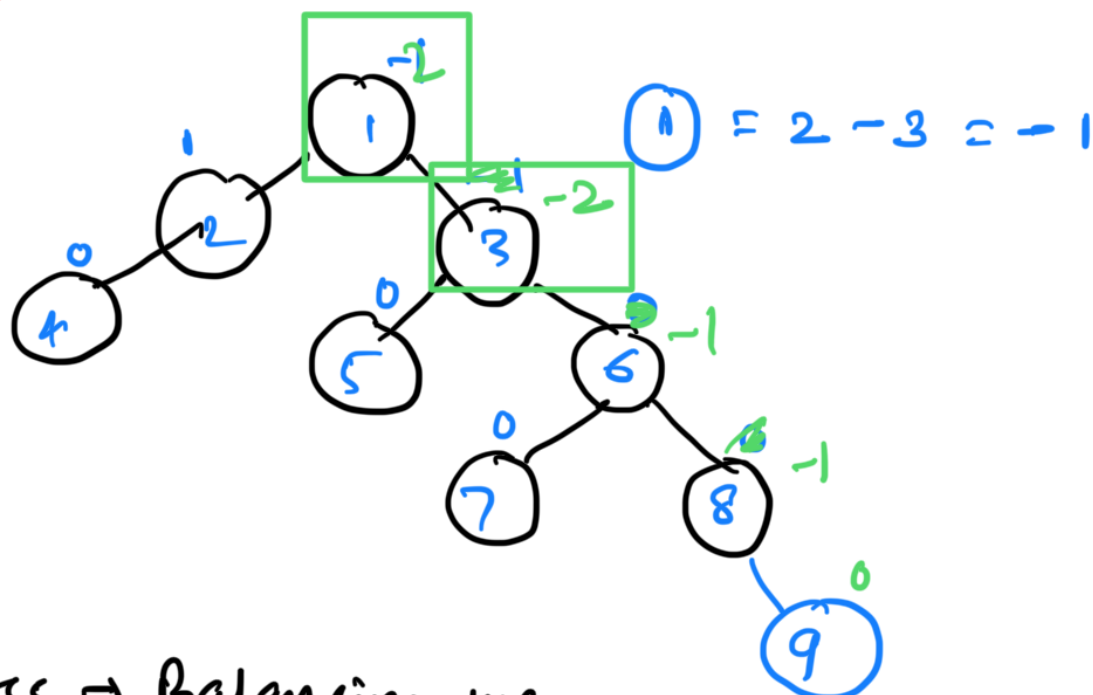
$$B = 0 - 1 = -1$$

$$A = 1 - 1 = 0$$

$$A = 2 - 0 = 2$$

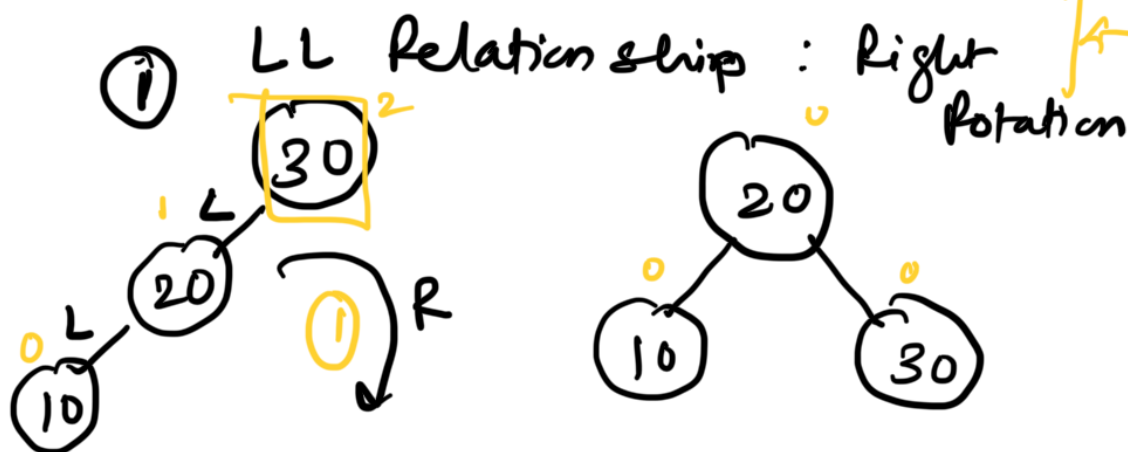
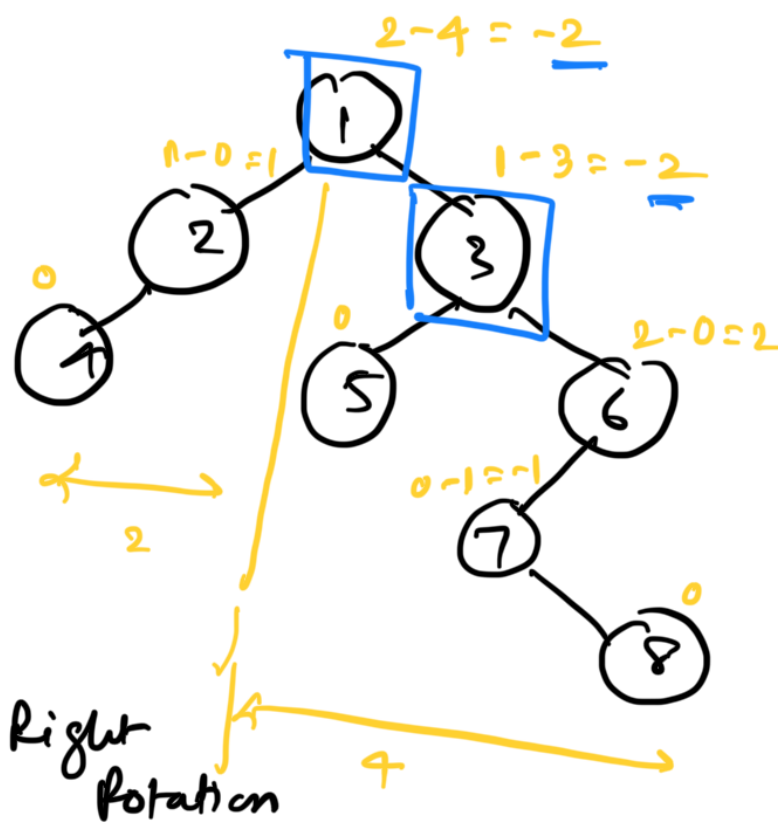
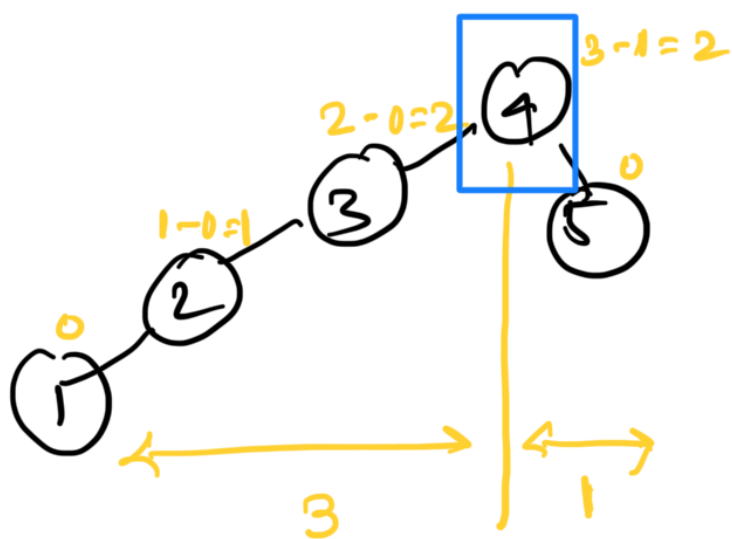
$$A = 0 - 2 = -2$$

Ex



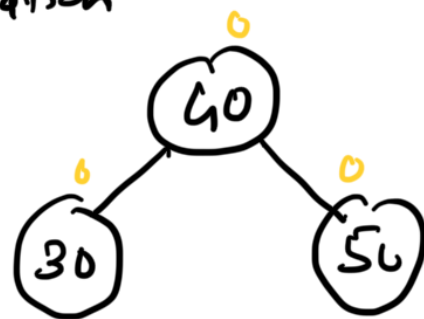
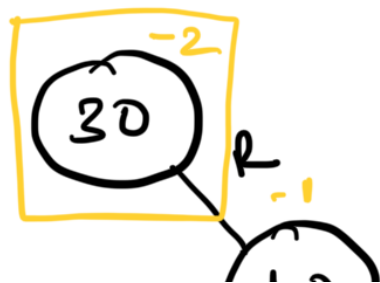
AVL Tree $\Rightarrow$ Balancing we use **ROTATIONS**

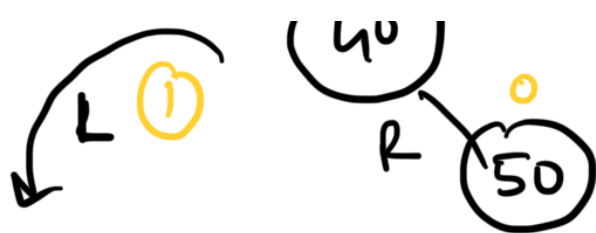
1. Single Rotation - L, R
 2. Double Rotation - L, R
- } 4 Rotations



① Right Rotation
LL

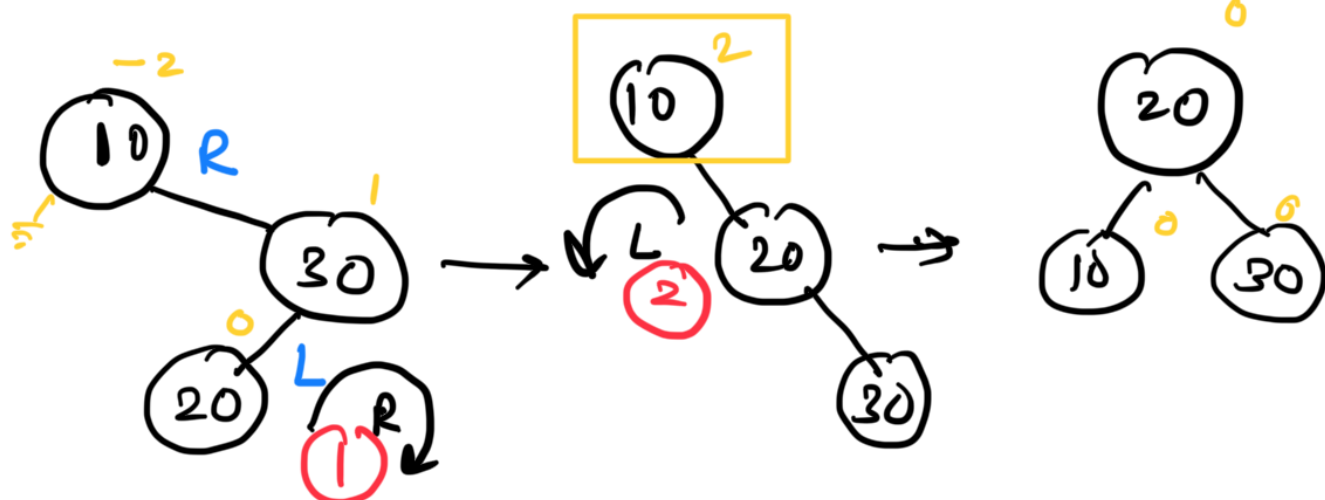
② RR Relationship : Left Rotation



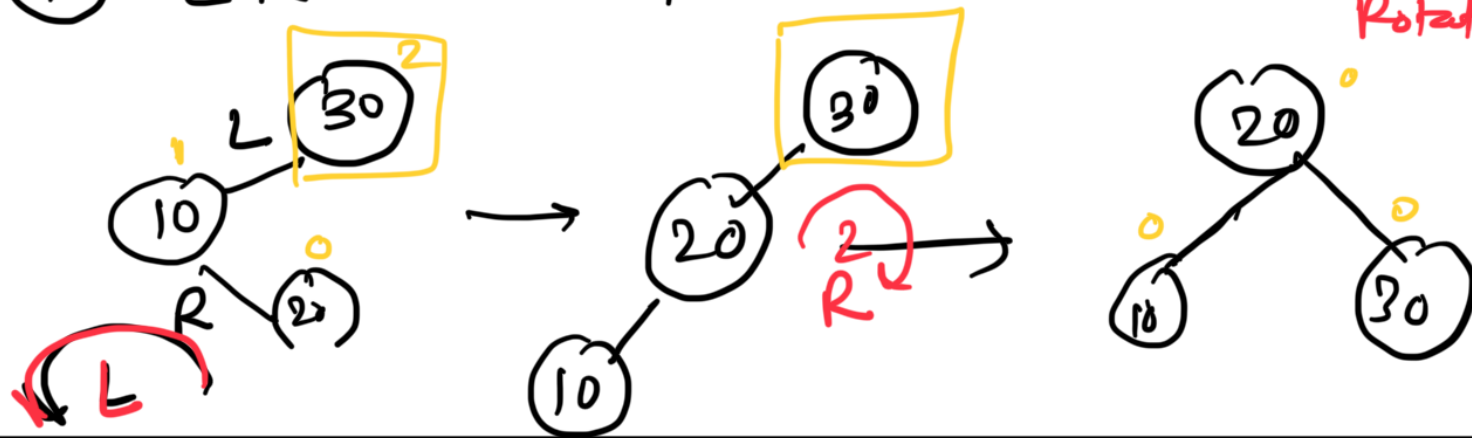


③ RL Relationship : RL Rotation

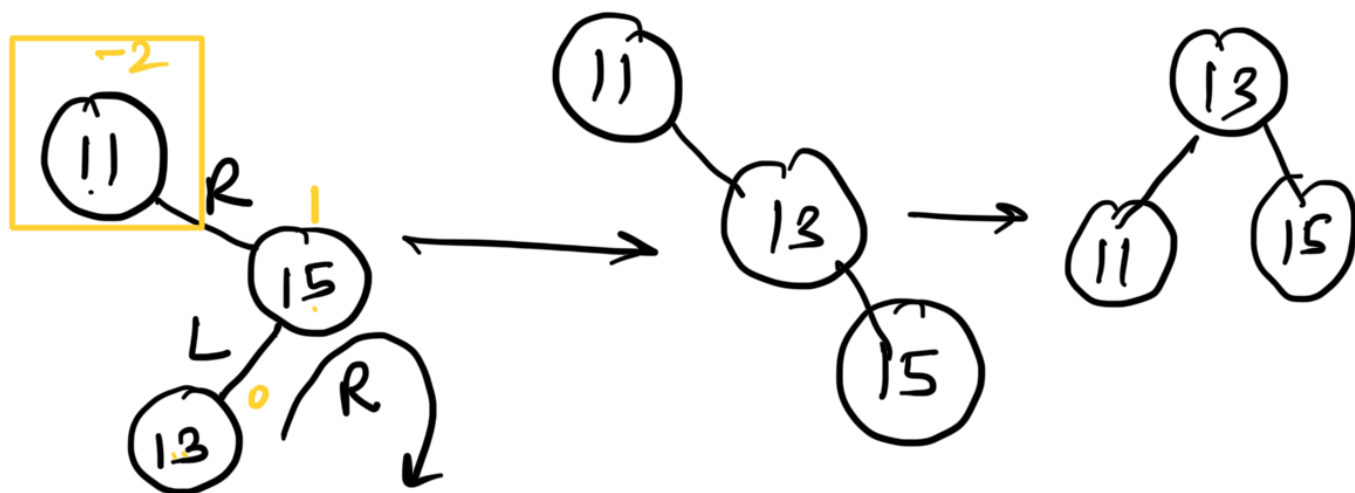
Double Rotation



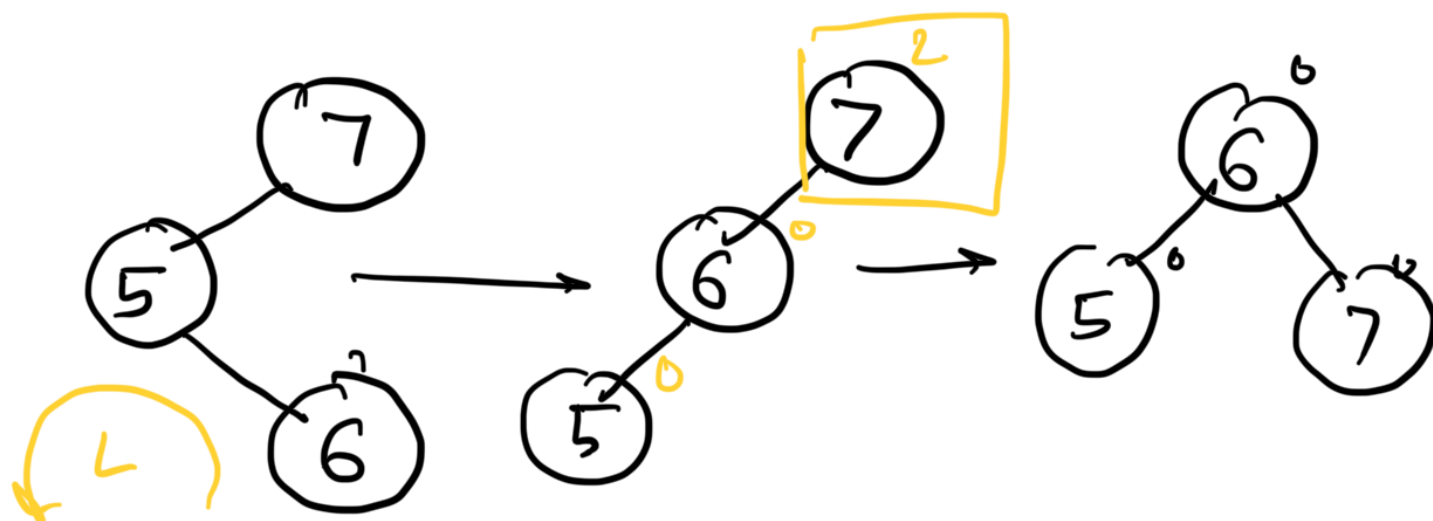
④ LR Relationship : LR Rotation - Double Rotation



Ex RL



Ex LR

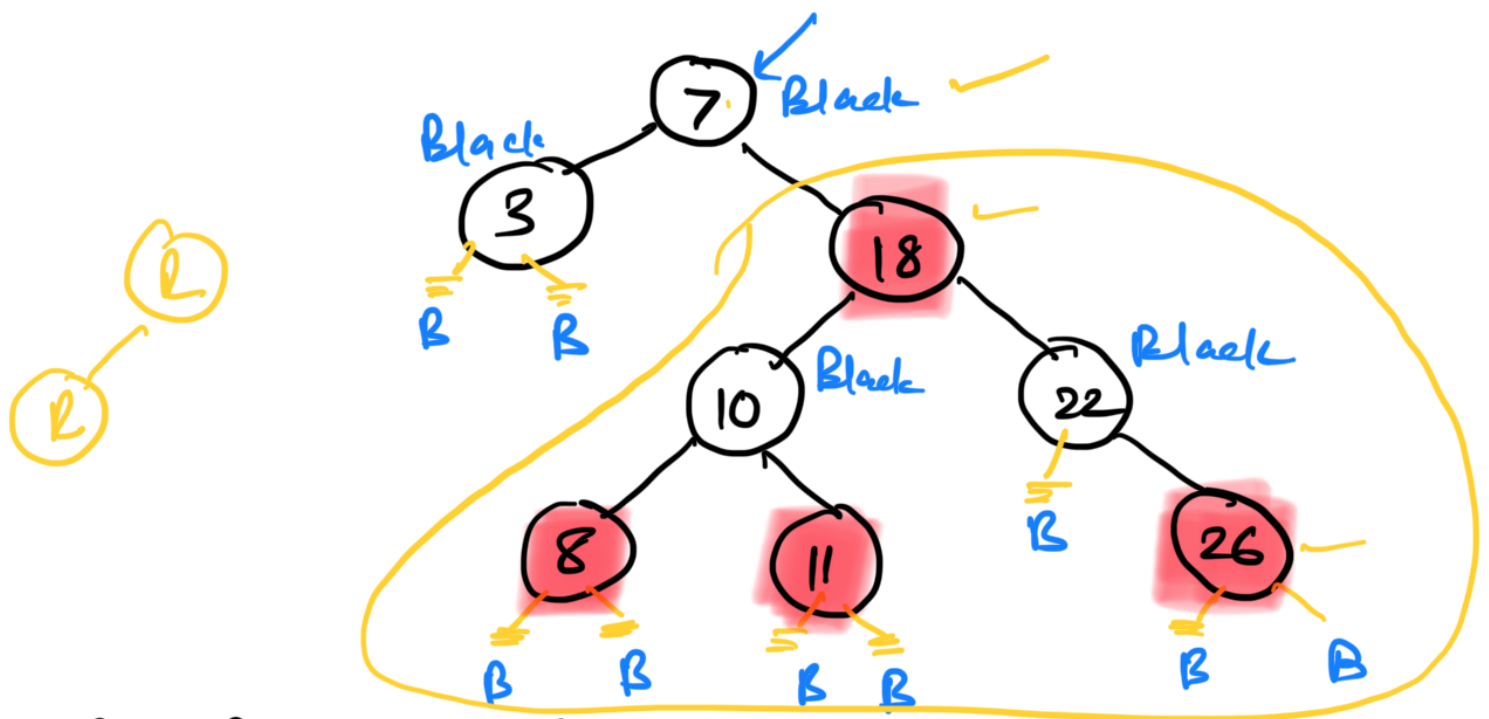


Red-Black Tree : BT = BST

color: Nodes → Red / Black

Balance → 1) color

2) Rotations - L, R, LR, RL



Red-Black Tree Properties

1. Root Property - The root is always black
2. Red Property - Red nodes cannot have red children (no two continuous red nodes)
3. Black Depth Property: Every path from a node to its descendant leaf has the same number of black nodes
4. Leaf property: Every leaf node (Null nodes) is black